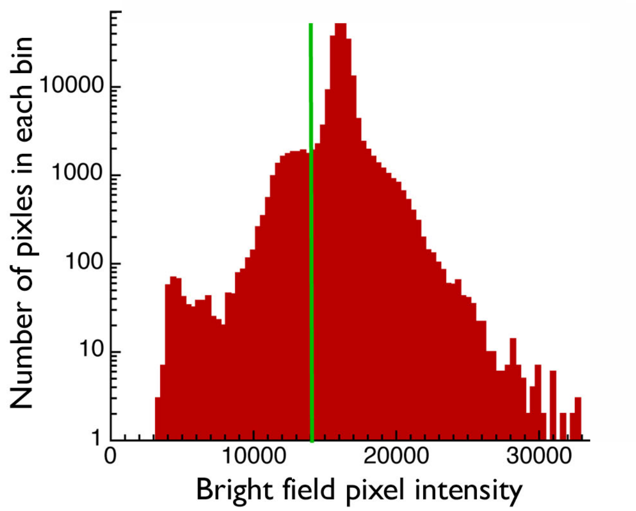
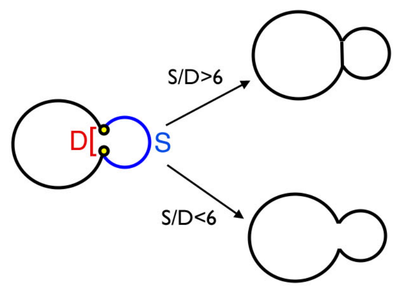
Supplementary Figure 1.
Examples of Cell-ID methodology

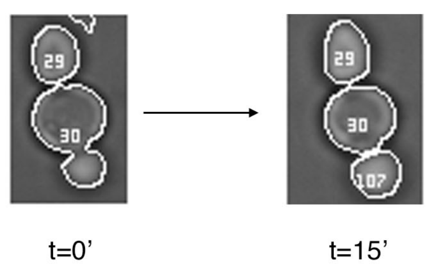
a



b



c



a. Histogram of the greyscale values of the 262,144 pixels that make up a representative 512x512 bright field image. Cell-ID looks for cell boundaries in the pixels whose greyscale values occur to left of the green line. **b.** Illustration of the calculation performed by Cell-ID on every cell to decide whether to split that cell into two cells. In this illustration Cell-ID has originally identified the single cell shown on the left. For every two points on the cell boundary, the program calculates the Euclidean distance between them (D) and also the smaller of the two distances to the left (black) and to the right (blue) along the cell boundary between them (S). It then finds the two points that have the maximal value for that cell for the ratio S/D. For a "figure eight" shape, these two points determine where the cell is most pinched (yellow points). If the ratio is higher than a user-defined limit (which defaults to 6), Cell-ID splits the original cells into two cells at the pinch (top). **c.** Example of a cell that gets split by the algorithm described in (b). Here we set the limit for S/D to be 8. Cell-ID calculated S/D=7.99 for cell #30 at T=0, and S/D=9.1 for the image taken fifteen minutes later. For the second image, Cell-ID gave the separated bud an ID number which had not yet been used by any cell at any previous time point, in this case #107.